

Design FMEA (qualitative version) — final design V8

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Qualitative ratings based on simulation risk signals only; this is a qualitative design FMEA, not a process/supplier FMEA.

#	Failure mode	Failure cause	Simulation signal (detectability basis)	Severity	Occurrence	Design-side mitigation
1	Lithium plating on negative electrode during 4C fast charge	anode interface saturation (non-uniform lithiation at high C-rate); aggravated by production drift of negative particle size / porosity	anode_potential_v min = +0.0119 V at 4C/45 °C (margin 11.9 mV) — rounds 1–3 showed crossings of –0.03 to –0.06 V before the transport package	High	Low	transport package implemented (neg r_p 2.0 μm, neg porosity 0.42, separator 9 μm/0.55, σ 0.5, σ 1.4 S/m); production-tolerance control on negative particle size and porosity; charge rate limited to 4C
2	Thermal limit exceeded (T > 60 °C)	heat generation > pack cooling under sustained load / higher ambient	T_max = 323.57 K at 4C/45 °C (9.6 K margin) and 308.45 K at 5C/25 °C	Medium	Low	h = 45 W/m²K ducted forced-air cooling assumed — pack must guarantee airflow; derate above 45 °C ambient
3	Electrolyte oxidative decomposition (voltage window)	upper cut-off overshoot / high-voltage surface states	voltage window 2.5–4.2 V respected in all protocols; molecular-level stability not endorsed (real_compute = false)	Medium	Low	charge cut-off fixed at 4.2 V (no overcharge protocol in use); Stage-5 DFT endorsement pending if real_compute is enabled
4	Insufficient capacity	electrode loading below tolerance / anode-limited balance drift	1C capacity 2.6683 Ah vs 2.0 Ah requirement (33 % margin)	Low	Low	electrode thickness/AMVF tolerance control; anode-limited design keeps capacity predictable (scales with negative AMVF, verified 0.4 %)
5	Internal short via separator defect	9 μm thin separator (power-cell class) with manufacturing defects	separator thickness 9 μm / porosity 0.55 (design values; defect statistics not simulable)	High	Low	ceramic-coated separator quality control; cell-level short-circuit screening on the production line

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6	Fast-charge aging acceleration	SEI growth / degradation under 4C cycling	not simulated in this case (aging model exists in parameter set but no aging protocol run)	Medium	Low	SEI-kinetics-capable parameter set (Chen2020) available for a follow-up aging study

Conclusion

Highest-risk items: #1 plating (thin +11.9 mV margin) and #5 thin separator — both mitigated in the design and flagged for production control. Complete process/supplier FMEA: N/A (beyond pure simulation boundary).